

PCB Design Workshop Activity

This workshop activity focuses on designing some of the power distribution as well as the motor driver circuitry for an RC car. This board acts as the bridge between your power source and the mechanical drive system. To gain practical experience, you may replicate the provided reference design or create a custom design, provided it meets the core requirements below.

1 Design Objectives

Your goal is to create a complete schematic and PCB layout for an RC car motor driver board. You are free to use KiCad, Altium Designer, or any professional EDA tool of your choice. Instructional videos and screenshots of the reference design are provided so you can follow along or copy the design exactly.

2 Core Schematic Requirements

Whether copying or customizing, your schematic must include the following functional blocks:

- **MCU Interfacing:** Pins or headers for a microcontroller (e.g., STM32 Nucleo or equivalent) to provide control signals.
- **Power Source:** The reference design is built for a 3S Lithium-Ion cell, but you may consider other battery configurations.
- **Voltage Regulation:** A circuit providing stable logic power (e.g., 5V regulated supply).
- **Motor Driver Circuit:** The reference design was for the PAN14EE12MD motor or a motor of your choice.
- **Protection Feature:** You must include at least one protection element (e.g., battery protection, reverse polarity, or fuses).
- **Peripherals:** Include status LEDs, test points, and standard mounting holes for the chassis.

3 Manufacturing & Layout Constraints

To ensure your board is compatible with standard low-cost manufacturing (such as JLCPCB's budget tier), you must adhere to the following physical constraints:

- **Board Dimensions:** The total board size must be a **maximum of 10 cm x 10 cm**.
- **Layer Count:** It must be a **2-layer board**.

Note: If you are planning to order the board on your own, you do not need to adhere to the 10 x 10 cm or 2-layer constraints. Please make note of this in your submission if you choose this path.

Additionally, the design should consider:

- **Trace Sizing:** Ensure power traces are wide enough to handle the current requirements of your chosen motor.
- **Mounting:** You must include the required mounting holes for your specific RC car chassis.

4 Recommended Tools & Reference BOM

To fast-track your design process, it is highly recommended to download and install the following tools:

- **Library Loader:** Download from [Component Search Engine](#) to easily import footprints and symbols directly into your EDA tool.
- **Saturn PCB Toolkit:** An excellent free resource for calculating required trace widths based on current and temperature rise.

5 Submission Instructions (Due Feb 1st)

Upload a **30-second "Fly-Through" screen recording** of your design to D2L.

- **Schematic (~10s):** Briefly show your wiring, labels, and protection features.
- **PCB Layout (~20s):** Show the 2D traces (Top and Bottom) and the 3D model view.
- **Verification:** Your name must be clearly visible in the **top right corner** of the software window during the recording.